WORKED SOLUTION

STEP III 1993, Question 16

Probability and Statistics • Truncated Normal Distribution

STEP III 1993, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q16

The key idea

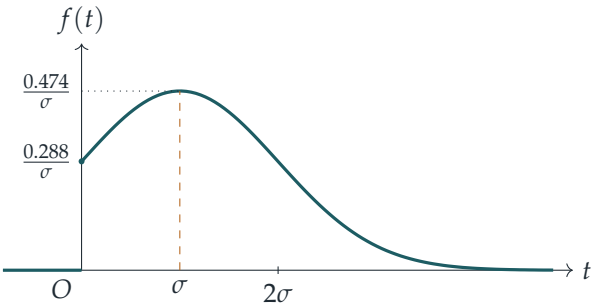
Apart from the factor $1/k$, the density is exactly that of a normal distribution with mean σ and standard deviation σ, cut off at $t = 0$. So every probability about T is a normal probability divided by the probability that the untruncated variable is non-negative. Standardising with $z = (t - \sigma)/\sigma$ turns everything into statements about $Z \sim \mathrm{N}(0,1)$ and the table of Φ.

Solution

Throughout, $Z \sim \mathrm{N}(0,1)$ with density $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ and distribution function Φ.

The sketch

For $t \geq 0$ the graph is the normal bell curve centred at $t = \sigma$, scaled up by $1/k$. It has its maximum at $t = \sigma$, where $f(\sigma) = \frac{1}{k\sigma\sqrt{2\pi}} \approx \frac{0.474}{\sigma}$, and a point of inflexion at $t = 2\sigma$. At the cut-off the curve starts at a positive height, $f(0) = \frac{e^{-1/2}}{k\sigma\sqrt{2\pi}} \approx \frac{0.288}{\sigma}$, and it jumps to 0 for $t < 0$. As $t \to \infty$, $f(t) \to 0$.



The value of k

The total probability must be 1:

$$1 = \int_0^\infty \frac{1}{k\sigma\sqrt{2\pi}} \exp\left(-\tfrac{1}{2}\left(\tfrac{t-\sigma}{\sigma}\right)^2\right) \mathrm{d}t.$$

Substituting $z = (t-\sigma)/\sigma$, so $\mathrm{d}t = \sigma\,\mathrm{d}z$ and $t = 0$ becomes $z = -1$,

$$1 = \frac{1}{k}\int_{-1}^\infty \phi(z)\,\mathrm{d}z = \frac{1-\Phi(-1)}{k} = \frac{\Phi(1)}{k}.$$

Hence $k = \Phi(1) \approx 0.8413$, so $k \approx 0.841$, as required.

The mean

With the same substitution, $t = \sigma(1+z)$, and

$$\mathrm{E}(T) = \frac{1}{k}\int_{-1}^\infty \sigma(1+z)\,\phi(z)\,\mathrm{d}z = \sigma\left[\frac{1}{k}\int_{-1}^\infty \phi(z)\,\mathrm{d}z + \frac{1}{k}\int_{-1}^\infty z\,\phi(z)\,\mathrm{d}z\right].$$

The first integral is k, as we have just shown.

Ask yourself How do we integrate $z\,\phi(z)$ when we can't integrate $\phi(z)$ itself?

We use the fact that $\phi'(z) = -z\,\phi(z)$, so $z\,\phi(z)$ has antiderivative $-\phi(z)$:

$$\int_{-1}^\infty z\,\phi(z)\,\mathrm{d}z = \Big[-\phi(z)\Big]_{-1}^\infty = \phi(-1) = \frac{e^{-1/2}}{\sqrt{2\pi}} \approx 0.24197.$$

Therefore

$$\mathrm{E}(T) = \sigma\left(1 + \frac{e^{-1/2}}{k\sqrt{2\pi}}\right) \approx \sigma\left(1 + \frac{0.24197}{0.84134}\right) \approx 1.288\,\sigma.$$

This is larger than σ, as it should be: removing the left tail pushes the mean to the right.

Finding σ

The mean setting time is 3 hours, so

$$\sigma = \frac{3}{1.2876} \approx 2.330 \text{ hours}.$$

For any $a \geq 0$, the same substitution gives

$$\mathrm{P}(T > a) = \frac{1}{k}\int_{(a-\sigma)/\sigma}^\infty \phi(z)\,\mathrm{d}z = \frac{1-\Phi\left(\frac{a}{\sigma}-1\right)}{\Phi(1)}. \qquad (*)$$

Part (i)

With $a = 4$: $\frac{4}{\sigma} - 1 \approx \frac{4}{2.330} - 1 \approx 0.717$, and $\Phi(0.717) \approx 0.7633$. By $(*)$,

$$\mathrm{P}(T > 4) \approx \frac{1-0.7633}{0.8413} \approx 0.2814, \qquad \mathrm{P}(T \leq 4) \approx 0.7186.$$

We now assume that the setting times of the 16 questions are independent, each with the distribution of T (the only reasonable model available). Then

$$\mathrm{P}(\text{no question takes more than 4 hours}) = \mathrm{P}(T \le 4)^{16} \approx 0.7186^{16} \approx 0.0051.$$

So it is very unlikely (about one chance in 200) that every question is set within 4 hours.

Part (ii)

Ask yourself Which event are we conditioning on?

"Unsatisfactory after 2 hours" means the question needs more than 2 hours in total, so it is the event $T > 2$. "Still unacceptable after a further 2 hours" is $T > 4$. Since $\{T > 4\} \subseteq \{T > 2\}$,

$$\mathrm{P}(T > 4 \mid T > 2) = \frac{\mathrm{P}(T > 4)}{\mathrm{P}(T > 2)}.$$

For $a = 2$: $\frac{2}{\sigma} - 1 \approx -0.142$, so $1 - \Phi(-0.142) = \Phi(0.142) \approx 0.5563$. In the ratio of the two expressions $(*)$ the factor $\Phi(1)$ cancels:

$$\mathrm{P}(T > 4 \mid T > 2) \approx \frac{1 - 0.7633}{0.5563} = \frac{0.2367}{0.5563} \approx 0.426.$$

Checks

- A direct numerical integration of the density (Simpson's rule, $\sigma = 1$) gives total area 0.84134 before dividing by k, and mean 1.2876σ.
- A simulation of 4×10^5 normal values with mean and standard deviation $\sigma = 2.330$, discarding the negative ones, gives sample mean 3.004, $\mathrm{P}(T > 4) \approx 0.282$ and $\mathrm{P}(T > 4 \mid T > 2) \approx 0.426$.
- Part (i) is very sensitive to rounding: $0.719^{16} \approx 0.0051$ but $0.718^{16} \approx 0.0050$. Quote it as about 0.005.
- Without the truncation, $\mathrm{P}(T > 4 \mid T > 2)$ would be the same, since the factor $1/k$ cancels. Only part (i) and the mean feel the effect of k.

Common slips

- Forgetting to divide by k in $\mathrm{P}(T > 4)$: this gives 0.237 instead of 0.281, and gives $0.7633^{16} \approx 0.013$ in (i), more than double the correct value.
- Taking the mean of T to be σ (the mean before truncation), which gives $\sigma = 3$.
- In (ii), finding $\mathrm{P}(2 < T \le 4)$ or $\mathrm{P}(T > 2)$ instead of the conditional probability.

Beyond STEP

The quantity $\phi(1)/\Phi(1)$ in the mean is an example of the *inverse Mills ratio*: in general a normal $\mathrm{N}(\mu, \sigma^2)$ truncated below at a has mean $\mu + \sigma\,\phi(\alpha)/(1-\Phi(\alpha))$ where $\alpha = (a-\mu)/\sigma$. Truncated distributions appear wherever data can only be observed above a threshold, for example in survival analysis and in econometric models of censored data. Part (ii) is about the *survival function* $\mathrm{P}(T > t)$; the exponential distribution is the one case where $\mathrm{P}(T > s+t \mid T > s)$ does not depend on s. Grimmett and Welsh, *Probability: An Introduction*, covers continuous distributions and conditioning.

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